THE STATE UNIVERSITY OF ZANZIBAR (SUZA)

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

SYSTEM DESIGN DOCUMENT (SDD)

DOCUMENT INFORMATION

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COURSE: Software Development Project

THE STATE UNIVERSITY OF ZANZIBAR (SUZA)

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1. INTRODUCTION

1.1 Purpose

The purpose of this System Design Document (SDD) is to describe the technical design of the **Citizen Alert System (CAS)**.This document explains **how the system will be designed and built**, including:

- System architecture
- Technology stack
- Database design
- API structure
- Security mechanisms
- User interface design

The document is intended for:

- Software developers
- Project supervisor
- ICT Department reviewers
- Technical stakeholders

It serves as a guide for implementing the Citizen Alert System based on the approved requirements.

1.2 Scope

This document covers the design of the complete Citizen Alert System, which is managed by the **Department of ICT - Zanzibar**. The scope includes:

A **mobile application** for citizens to receive official government alerts

A **web-based admin panel** for government offices

A **super admin panel** for the ICT Department

- ✓ Backend services for authentication, notification targeting, and data processing
- ✓ Database design for users, offices, locations, and notifications
- ✓ Integration with push notification services (e.g., Firebase Cloud Messaging)

This document does **not** cover:

- ✓ Legal policy documents
- ✓ Project budgeting
- ✓ Detailed testing plans

1.3 Design Goals

The main design goals of the Citizen Alert System are:

Centralization:
Provide a single, official platform for all government alerts.

Scalability:
Support a growing number of citizens, offices, and notifications.

Security:
Protect user data and ensure only verified offices can send alerts.

Performance:
Deliver notifications quickly, even during emergencies.

Relevance:
Send alerts only to the right users based on location and profile.

Usability:
Ensure the system is easy to use for citizens and government staff.

2. SYSTEM ARCHITECTURE

2.1 Architecture Overview

The Citizen Alert System (CAS) uses a **centralized client–server architecture** managed by the **Department of ICT – Zanzibar**.

The system is made up of four main parts:

Mobile Application (Citizen App)

- Used by citizens to receive alerts and announcements
- Built using React Native
- Runs on Android (and iOS in the future)

Web Admin Panel (Government Offices)

- Used by ministries and government offices (clients)
- Built using React Web
- Allows offices to create and manage notifications

Super Admin Panel (ICT Department)

- Used by the ICT Department
- Manages offices, approvals, policies, and system control

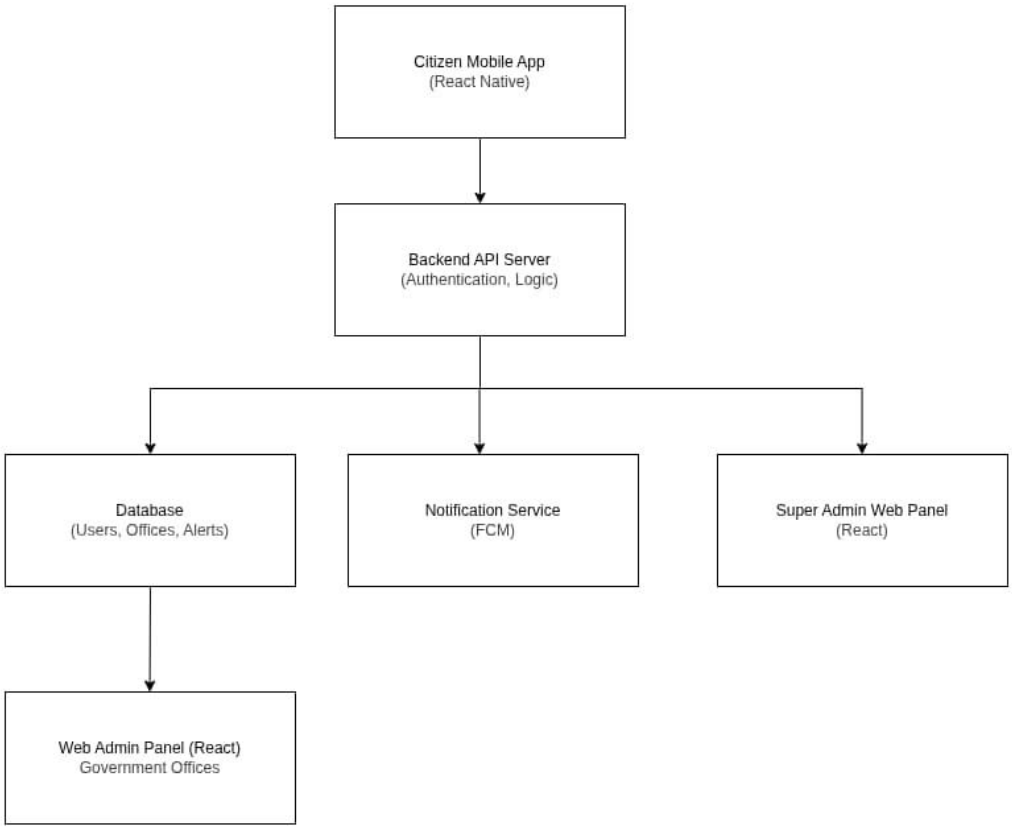
Backend Server and Database

- Handles authentication, notification targeting, and data storage
- Acts as the central brain of the system

All clients communicate with the backend through secure internet connections.

2.2 Architecture Diagram

The high-level architecture of CAS is shown below:



2.3 Architecture Pattern

Pattern Used: Layered (Three-Tier Architecture)

Architecture Layers

- **Presentation Layer**

  Citizen Mobile App

  Office Admin Web Panel

  ICT Super Admin Panel

- **Application Layer** Backend API Server

  Business logic

  Notification targeting rules

- **Data Layer**

  Relational database (PostgreSQL / MySQL)

3. TECHNOLOGY STACK

3.1 Frontend Technologies

Technology	Version	Purpose
React Native	Latest stable	Mobile application for citizens
React.js	Latest stable	Web admin panel for offices and ICT
HTML5	5	Page structure
CSS3 / Tailwind CSS	Latest	Styling and responsive design
JavaScript / TypeScript	Latest	Client-side logic

3.2 Backend Technologies

Technology	Version	Purpose
Node.js	Latest LTS	Server runtime environment
Express.js	Latest	Backend web framework
JWT	Latest	Authentication and authorization
Firebase Admin SDK	Latest	Notification delivery service

3.3 Database

Technology	Type	
PostgreSQL / MySQL	Relational	Main system database
Redis (Optional)	In-memory	Caching and performance
Firebase Cloud Messaging (FCM)	Notification	Push notifications

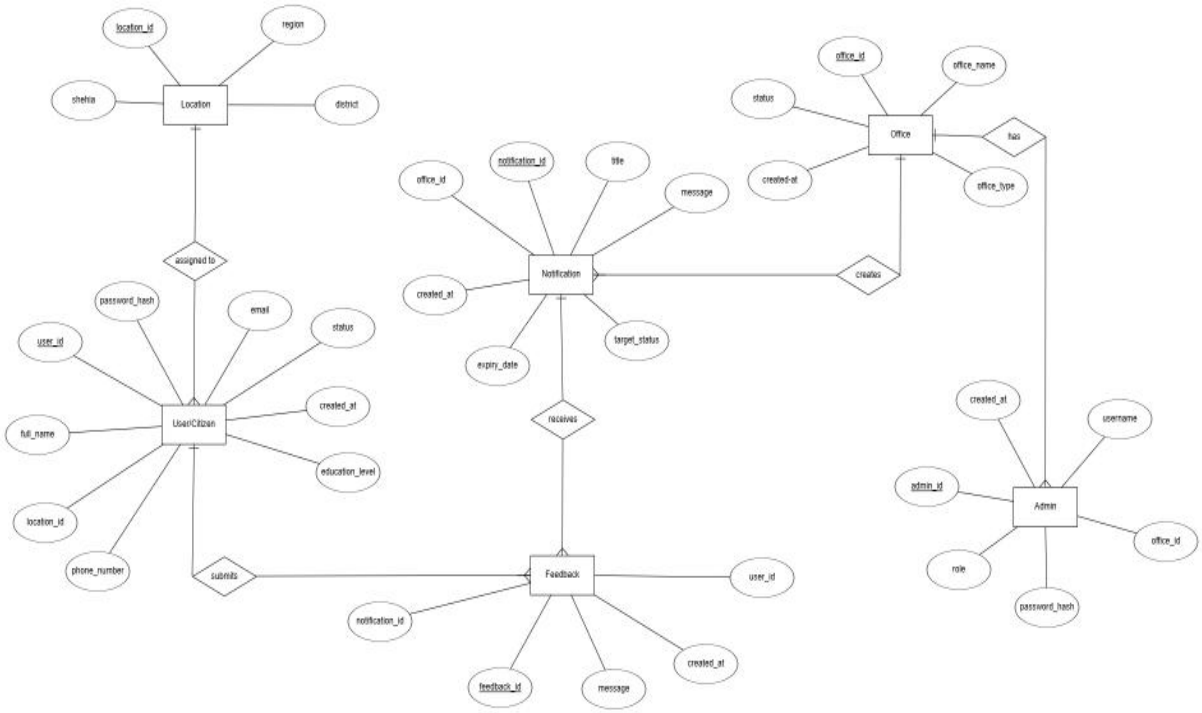
3.4 DevOps and Deployment

Tool	Purpose
Git & GitHub	Version control
Docker	Containerization
Government Data Center / Cloud	System hosting
Nginx	Reverse proxy and security
CI/CD (GitHub Actions)	Automated deployment

4. DATABASE DESIGN

4.1 Entity Relationship Diagram (ERD)

The Citizen Alert System database is designed to store users, government offices, notifications, locations, and feedback.



5. API DESIGN

5.1 API Overview

The CAS backend uses **RESTful APIs** to allow communication between:

Citizen Mobile App (React Native)

Office Admin Web Panel (React Web)

Super Admin Panel (React Web)

The APIs handle:

Authentication and authorization

User registration and login

Notification creation, targeting, and retrieval

Feedback submission and retrieval

Office and admin management

All communication is **secured with HTTPS** and **JWT tokens** for authenticated users.

5.2 API Endpoints

Authentication Endpoints

Method	Endpoint	Description	Auth Required
POST	/api/auth/register	Citizen registration	No
POST	/api/auth/login	User login (Citizen/Admin)	No
POST	/api/auth/logout	Logout	Yes

Citizen Resource Endpoints

Method	Endpoint	Description	Auth Required
GET	/api/notifications	Get notifications for citizen	Yes
GET	/api/notifications/:id	Get notification dedetails	Yes
POST	/api/feedback	Submit feedback	Yes

Office Admin Endpoints

Method	Endpoint	Description	Auth Required
POST	/api/notifications	Create new notification	**Yes**
GET	/api/notifications	List office notifications	**Yes**
PUT	/api/notifications/:id	Update notification	**Yes**
DELETE	/api/notifications/:id	Delete notification	**Yes**

Super Admin Endpoints (ICT Department)

Method	Endpoint	Description	Auth Required
GET	/api/offices	List all offices	Yes
POST	/api/offices/approve	Approve office registration	Yes
PUT	/api/offices/:id	Update office details	Yes
DELETE	/api/offices/:id	Suspend or remove office	Yes
GET	/api/users	List all users	Yes

5.3 Request / Response Examples

1. Citizen Registration

Request (POST /api/auth/register):

```
{
  "full_name": "Khadija Said",
  "phone_number": "255712345678",
  "email": "khalsa@gmail.com",
  "password": "SecurePass123",
  "status": "Student",
  "education_level": "Undergraduate",
  "location_id": 2
}
```

Response (Success - 201):

```
{
  "success": true,
  "message": "User registered successfully",
  "data": {
    "user_id": 101,
    "full_name": "Khadija Said",
    "phone_number": "255712345678"
  }
}
```

2. Create Notification (Office Admin)

Request (POST /api/notifications):

```
{
  "title": "Power Outage in Bububu",
  "message": "Planned electricity outage from 9 AM to 1 PM",
  "category": "Public Services",
  "target_status": "Citizen",
  "target_education": null,
  "location_id": 2,
  "expiry_date": "2026-02-15"
}
```

Response (Success - 201):

```
{
  "success": true,
  "message": "Notification created successfully",
  "data": {
    "notification_id": 501,
    "title": "Power Outage in Bububu"
  }
}
```

This API design **covers all main interactions** for citizens, office admins, and the ICT super admin.

6. USER INTERFACE DESIGN

6.1 UI/UX Principles

The CAS UI is designed for **ease of use**, **clarity**, and **accessibility**:

Consistency: Same style, buttons, and colors across all screens

Mobile-first design: Optimized for Android phones, responsive for iOS

Clear navigation: Simple menus and back buttons

Feedback: Users see confirmations, errors, and notifications

Multilingual support: Swahili and English

6.2 Screen Layouts

1. Citizen Mobile App

Login / Registration Screen

Fields: Phone number, password, OTP verification

Buttons: Login, Register, Forgot Password

Home / Notification Feed

List of notifications relevant to the citizen

Filters: Category, Location, Status

Indicators: Read / Unread notifications

Notification Details

Title, message, category, office name, expiry date

Button: “Submit Feedback”

Settings / Profile

Update location, status, education level

Change language

2. Office Admin Web Panel

Dashboard

Quick view: Pending approvals, sent notifications, feedback summary

Create Notification Screen

Fields: Title, Message, Category, Target Location, Target Status, Expiry Date

Button: Submit for approval

Notification Management

Edit, Delete, View history of notifications

Reports / Analytics

Delivery reports, read/unread status

3. Super Admin Web Panel (ICT Department)

Admin Dashboard

Approve or reject offices

View all notifications and feedback

Manage system policies and users

Office Management

Add, suspend, or remove offices

Track office performance

6.3 Navigation Flow

Citizen App

Login / Register --> Home Feed --> Notification Details --> Submit Feedback|--> Filters / Settings

Office Admin Panel

Login --> Dashboard --> Create Notification --> Manage Notifications --> Reports

Super Admin Panel

Login --> Dashboard --> Manage Offices--> View Notifications--> Manage Users / Policies

7. SECURITY DESIGN

7.1 Authentication

Method: JWT (JSON Web Tokens)

How it works:

Users (citizens, admins) login with credentials

System verifies credentials and issues a **token**

Token is sent with every API request

Token Expiration: 30 minutes

Refresh Token: Yes, for session continuation without re-login

Additional Security for Admins: Two-factor authentication (2FA)

7.2 Authorization

Role-based Access Control (RBAC):

Super Admin (ICT Department): Full access

Office Admin: Access only to their office's notifications

Citizen: Access only to notifications targeted to them

Permissions are checked at API level to prevent unauthorized access

7.3 Data Protection

Passwords:

Stored using **bcrypt hashing**

Never stored in plain text

Communication:

All API calls over **HTTPS**

Prevents data interception

Input Validation & Sanitization:

Prevent SQL injection

Prevent XSS attacks (malicious scripts)

Database Security:

Restricted access per role

Daily backups for reliability

Notification Security:

Push notifications sent securely via **Firebase Cloud Messaging (FCM)**

Only targeted users receive alerts

This design ensures **citizen data and government notifications are safe, private, and reliable**.

8. COMPONENT DESIGN

8.1 Component 1: User Module (Citizen Mobile App)

Purpose: Handles all citizen interactions

Responsibilities:

Register and login

Receive notifications

View notification history

Submit feedback

Manage profile and settings

Dependencies: Backend APIs, Firebase Cloud Messaging, Local storage

Interface: API calls for authentication, notifications, feedback

8.2 Component 2: Notification Module

Purpose: Manages creation, delivery, and targeting of notifications

Responsibilities:

Create, update, delete notifications (Office Admin)

Target users based on location, status, education level

Track read/unread status

Schedule and expire notifications

Dependencies: User module, Office module, FCM service

Interface: REST API endpoints for creating and fetching notifications

8.3 Component 3: Feedback Module

Purpose: Handles citizen feedback on notifications

Responsibilities:

Receive feedback submissions

Notify admins of new feedback

Store feedback in database

Dependencies: User module, Notification module, Admin module

Interface: API endpoints for feedback submission and retrieval

8.4 Component 4: Admin Module (Web Dashboard)

Purpose: Allows government offices to manage notifications

Responsibilities:

Login and authentication

Create and manage notifications

View reports and feedback

Dependencies: Notification module, Feedback module

Interface: Admin web panel UI connecting to backend APIs

8.5 Component 5: Super Admin Module

Purpose: ICT Department control and oversight

Responsibilities:

Approve/reject office registration

Manage system users

View all notifications and feedback

Configure system rules

Dependencies: Office module, Admin module, Notification module

Interface: Super Admin web panel

This **modular design** makes the system **scalable, maintainable, and secure**.

9. DEPLOYMENT ARCHITECTURE

The **deployment architecture** explains how the Citizen Alert System (CAS) components are **hosted, connected, and interact** in real life. It ensures the system is **accessible, reliable, and secure** for citizens, office admins, and ICT super admins.

9.1 Deployment explanation

- **Citizen Mobile App:**

Receives notifications, views announcements, and submits feedback.

- **Web Admin Panel:**

Office admins create and manage notifications. Super admins manage offices and system policies.

- **Backend Server:**

Handles authentication, authorization, business logic, and API requests.

- **Database:**

Stores all system data securely (users, notifications, feedback, offices, locations).

- - **Firebase Cloud Messaging (FCM):**

Sends push notifications to citizens in real-time based on filters (location, status, education).

9.2 Environment Configuration

Development Environment:

Runs locally on developer machines (localhost)

Backend, database, and notification service are tested here

Purpose: Safe environment for coding and initial testing

Staging Environment:

URL: staging.cas.zanzibar

Mirrors the production setup for testing

Purpose: QA, integration testing, and load testing before live release

Production Environment:

URL: www.cas.zanzibar

Live system used by citizens and government offices

Backend and database hosted on secure servers or cloud

Push notifications delivered through FCM

HTTPS enforced for all communication

Daily backups and monitoring implemented

10. APPENDICES

10.1 Glossary

CAS (Citizen Alert System): The digital platform for official government notifications in Zanzibar.

ICT Department: Government authority managing CAS, approving offices, and ensuring security.

Citizen/User: A registered person receiving notifications via the mobile app.

Office Admin: Authorized staff from government offices who create and manage notifications.

Super Admin: ICT Department official with full access to the system.

Notification: An official message sent to targeted citizens based on location, status, or education level.

Feedback: Comments or reports submitted by citizens about a notification.

Location: Geographic information (Region → District → Shehia) used to target notifications.

JWT (JSON Web Token): Token used for secure authentication of users.

FCM (Firebase Cloud Messaging): Service used to send push notifications to citizens' devices.

10.2 Revision History

Version 1.0 - February 2026

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Initial draft of System Design Document (SDD)

Includes: System Architecture, Technology Stack, Database Design, API Design, UI Design, Security Design, Component Design, Deployment Architecture, Appendices